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Comparison of machine learning methods for the task of predicting
antimicrobial properties of peptides

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Chapter 1

Introduction

At present, the number of infections with drug-resistant bacteria is increasing, to which Conventional antibiotics are becoming less and less effective. This creates a risk of a world without antibiotics, which until now have been effective measures against bacterial infections. For this reason, more and more attention is being paid to unconventional antibacterial agents, including antimicrobial peptides (AMPs), due to their potential as a new class of compounds enabling the fight against pathogenic microorganisms.[1]

1.1 Peptides

Peptides are short chains of amino acids linked by peptide bonds, generally defined as molecules composed of between two and fifty amino acids [2]. They occupy a unique space in the biomolecule hierarchy, bridging the structural and functional gap between small molecules and large proteins. The distinction between peptides and proteins is largely a matter of size and complexity: proteins typically consist of one or more polypeptide chains with well-defined three-dimensional structures, while peptides are shorter, more flexible and often lack a stable tertiary conformation [3]. This structural difference has profound implications for their biological roles and therapeutic applications.

Functionally, peptides differ from proteins in that they frequently act as regulators rather than structural or catalytic components of the cell. Many naturally occurring peptides serve as hormones, neurotransmitters, and signaling molecules, exerting precise control over physiological processes such as glucose regulation, pain perception, and immune modulation [4]. Their relatively small size allows them to interact with receptors and enzymes with high specificity, often mimicking endogenous ligands or modulating protein-protein interactions that are otherwise inaccessible to conventional small molecules. This regulatory nature is one reason why peptides are increasingly investigated as drug candidates.

From a biomedical perspective, peptides represent an important therapeutic class located between traditional small molecules and large biologics. Small molecules often suffer

from low selectivity, while large molecules such as monoclonal antibodies can face delivery challenges. However, peptides combine advantages from both worlds: they exhibit high selectivity and potency, reduced toxicity, and due to their smaller size compared to proteins, often show improved tissue penetration [2]. At the same time, they are more stable and less immunogenic than many protein-based drugs, making them particularly attractive for clinical development.

Despite these advantages, peptides also face limitations. Their susceptibility to enzymatic degradation, short circulation half-life, and poor oral bioavailability present challenges to drug design and delivery (Otvos, 2016). To overcome these barriers, a variety of strategies have been developed, such as cyclization, incorporation of non-natural amino acids, and conjugation with polymers (e.g., PEGylation) [4]. These innovations not only improve peptide stability but also expand their pharmacological potential.

The growing interest in peptides is reflected in the pharmaceutical market, where peptide-based therapeutics are increasingly common. Examples include insulin, a peptide hormone regulating blood glucose; glucagon-like peptide-1 (GLP-1) analogues for diabetes management; and enfuvirtide, an antiviral peptide used in HIV therapy. These successes highlight how peptides, despite their limitations, can be optimized into clinically relevant drugs.

In the context of antimicrobial peptides (AMPs), peptides' regulatory and signaling functions take on an additional dimension. AMPs combine structural simplicity with potent biological activity, and their short sequence length allows rapid evolution and adaptation across species. This makes peptides an invaluable subject of study not only for basic biological insight but also for the development of novel therapeutic strategies, particularly in an age of increasing antimicrobial resistance.

1.2 Antimicrobial Peptides (AMP)

Antimicrobial peptides (AMPs) are a diverse class of short, naturally occurring peptides that form a crucial part of the innate immune system across virtually all forms of life. Typically consisting of 10–50 amino acids, AMPs exhibit broad-spectrum activity against bacteria, fungi, viruses, and even malignant cells [5]. Their importance stems from their evolutionary role as the first line of defense against microbial invasion, complementing more specialized immune responses in higher organisms. Unlike conventional antibiotics that often target specific metabolic pathways, AMPs exert their activity primarily through physical disruption of microbial membranes, making them less prone to resistance development[6].

One of the defining features of AMPs is their structural diversity. Despite their short length, they frequently adopt amphipathic conformations with distinct hydrophobic and cationic regions, allowing interaction with negatively charged microbial membranes [7].

Upon contact, many AMPs insert themselves into lipid bilayers, causing destabilization and pore formation that lead to rapid cell death. Other AMPs exert nonmembranolytic effects by interfering with intracellular targets such as nucleic acids, protein synthesis, or enzymatic activity. Some AMPs are also known to modulate host immune responses, acting as signaling molecules that recruit immune cells and promote wound healing [8].

The range of biological targets that AMPs affect is broad. Although many exhibit strong bactericidal activity against both Gram-positive and Gram-negative species, others display potent antifungal activity, antiviral effects, or cytotoxicity toward cancer cells [9]. This functional diversity has led to growing interest in AMPs as templates for novel therapeutic agents, especially in light of the global threat posed by antimicrobial resistance. Traditional antibiotics are losing their effectiveness at an alarming rate due to the evolution of resistant strains, but the multitarget mechanisms of AMPs reduce the likelihood that resistance arises as quickly [6].

However, AMPs are not without challenges. Many naturally occurring AMPs are cytotoxic to host cells at higher concentrations and their stability in physiological environments can be limited due to proteolytic degradation. In addition, their relatively short half-lives and high production costs have hindered widespread clinical adoption (Mansour et al., 2014). Strategies to overcome these limitations include chemical modifications such as peptide cyclization, incorporation of D-amino acids, and synthetic optimization of AMP analogs to improve selectivity and stability [9].

From an evolutionary perspective, AMPs are particularly fascinating because they have arisen independently in a wide range of organisms, from plants and insects to amphibians and mammals. Databases such as the Antimicrobial Peptide Database (APD), the Database of Antimicrobial Activity and Structure of Peptides (DBAASP), and the Data Repository of Antimicrobial Peptides (DRAMP) catalog thousands of naturally occurring and synthetic AMPs, underscoring their abundance and diversity. These repositories are critical resources for computational prediction and machine learning applications, as they provide large curated datasets linking peptide sequences to their biological activities.

Given their versatility and therapeutic promise, AMPs represent a unique frontier in drug discovery. Their multifunctional nature, simultaneously antimicrobial, immunomodulatory, and occasionally anticancer, differentiates them from conventional drugs. For this reason, computational methods aimed at predicting AMP function, such as those developed in this thesis, are not only academically relevant but also practically significant for guiding the design of next-generation peptide-based therapeutics.

1.3 Exemple sources of AMPs

Antimicrobial peptides are evolutionarily conserved molecules that have been identified in virtually all living organisms, from microorganisms to humans. Their ubiquity reflects

their crucial role in innate defense systems across the tree of life. Because AMPs are small, versatile and inexpensive to produce, they serve as effective first-line defense mechanisms against microbial infections in a wide range of environments [10].

Bacterial sources. Certain bacteria produce AMPs, known as bacteriocins, which provide competitive advantages by inhibiting the growth of rival species. Bacteriocins are highly diverse in structure and mode of action, ranging from pore-forming peptides to enzymes that degrade cell walls [11]. Examples include nisin, a lantibiotic produced by *Lactococcus lactis*, which has been used commercially as a food preservative due to its effectiveness against Gram-positive bacteria.

Plant-derived AMPs. Plants have evolved a wide variety of AMPs as part of their chemical arsenal against pathogens and herbivores. Plant defensins, thionins, and cyclotides are among the best-studied families. These peptides protect plants by disrupting microbial membranes or interfering with essential intracellular processes [12]. Cyclotides, for instance, are cyclic peptides found in certain plant families such as Rubiaceae and Violaceae, characterized by exceptional structural stability that makes them attractive scaffolds for drug design.

Invertebrate sources. Arthropods, such as insects, produce AMPs as part of their hemolymph defense. Drosomycin from *Drosophila melanogaster* and defensins from mosquitoes are classical examples that highlight the centrality of AMPs in insect immunity [13]. The evolutionary success of insects has in part been attributed to their ability to rapidly deploy diverse AMP repertoires against bacterial and fungal pathogens.

Amphibians. Amphibian skin is one of the richest natural reservoirs of AMPs, with species such as frogs secreting a wide array of peptides to protect against microbial colonization in their moist habitats [14]. Magainins, first isolated from the skin of *Xenopus laevis*, represent one of the earliest discovered amphibian AMPs and have inspired numerous synthetic derivatives with therapeutic potential.

Mammalian sources. In humans and other mammals, AMPs form an essential component of innate immunity. The most prominent families include defensins and cathelicidins, which are expressed in epithelial tissues, neutrophils, and mucosal surfaces. Human cathelicidin LL-37 not only exhibits potent antimicrobial activity but also modulates inflammatory responses, highlighting the dual protective and regulatory roles of mammalian AMPs [15]. These peptides contribute to host defense against a wide range of pathogens, from Gram-positive and Gram-negative bacteria to fungi and viruses.

Marine organisms. More recently, AMPs have been isolated from marine organisms, including fish, mollusks, and crustaceans. These peptides are of particular interest due to their novel structures and mechanisms of action shaped by unique marine environments [16]. Piscidins, for instance, are a class of AMPs from fish that display broad-spectrum antimicrobial and immunomodulatory activities.

Collectively, these examples highlight the evolutionary convergence toward peptide-

based defense strategies across kingdoms. This diversity not only illustrates the fundamental role of AMPs in survival but also provides a vast reservoir of templates for therapeutic development. For computational studies, the cataloging of AMPs from such diverse sources into curated databases provides the essential raw material for building predictive models, including the machine learning approaches used in this thesis.

1.4 Sequencing

The identification and characterization of antimicrobial peptides (AMPs) rely heavily on sequencing technologies, which provide insights into their amino acid composition, structure, and functional potential. Sequencing enables researchers to uncover novel peptides from natural sources and annotate them in specialized databases, forming the foundation for computational modeling and predictive tasks such as those undertaken in this thesis. Over the past decades, peptide sequencing has undergone a major transformation—from labor-intensive classical methods to modern high-throughput technologies.

Edman degradation. Historically, the first reliable peptide sequencing technique was Edman degradation, developed in the 1950s. This method involves sequentially labeling and cleaving the N-terminal residue of a peptide, followed by chromatographic identification [17]. While highly accurate for short peptides, Edman degradation is limited to sequences of 30–40 residues and requires substantial amounts of purified peptide, making it impractical for complex AMP discovery workflows.

Mass spectrometry-based sequencing. The development of modern mass spectrometry (MS) has revolutionized peptide sequencing. Techniques such as matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI), combined with tandem MS (MS/MS), allow for rapid *de novo* sequencing of peptides, even from complex biological mixtures [18]. Mass spectrometry is particularly valuable in AMP research, as many peptides are small, cationic, and structurally diverse, often not directly predictable from genomic data. Moreover, MS can reveal post-translational modifications, which play critical roles in the bioactivity of AMPs, such as cyclization or disulfide bond formation.

Sanger sequencing of DNA. While not directly sequencing peptides, Sanger sequencing of DNA provided a crucial step toward identifying AMPs by revealing the genes encoding precursor proteins. In the 1970s, Sanger sequencing laid the foundation for gene-based peptide discovery, particularly for AMPs stored in precursor proteins with signal sequences [19]. However, its low throughput has since limited its application in large-scale AMP exploration.

Next-generation sequencing (NGS). The advent of next-generation sequencing (NGS) has greatly expanded the scope of AMP discovery. NGS enables the rapid sequencing of entire genomes and transcriptomes, providing indirect but powerful access to AMP-encoding genes [20]. Transcriptomic approaches, in particular, have facilitated the identification of

inducible AMPs expressed in response to infection or environmental stress. Combined with bioinformatics pipelines, NGS data allow for the large-scale prediction of peptide sequences that can then be validated experimentally.

Single-cell and metagenomic sequencing. More recently, single-cell sequencing and metagenomics have been employed to explore unculturable organisms and complex microbiomes as potential AMP sources [21]. These approaches bypass cultivation bottlenecks, unlocking vast reservoirs of genetic diversity from which new AMPs can be mined.

Together, these sequencing technologies provide complementary perspectives: classical Edman degradation and mass spectrometry offer direct insight into peptide chemistry, while DNA- and RNA-based sequencing approaches provide access to the genetic blueprints of peptides across diverse organisms. For computational AMP research, the integration of sequence data from mass spectrometry and NGS into curated databases such as DBAASP, DRAMP, and APD has been indispensable. These databases serve as both repositories and training grounds for predictive machine learning models, including those applied in this study.

1.5 AMP databases

The rapid progress in antimicrobial peptide (AMP) research has been accompanied by the development of dedicated databases that systematically collect, annotate, and distribute peptide sequence and activity information. These databases are essential for both experimental and computational studies, as they provide curated resources that enable large-scale statistical analysis, benchmarking, and machine learning (ML) applications. In particular, three databases have emerged as the most widely used and influential: the Database of Antimicrobial Activity and Structure of Peptides (DBAASP), the Database of Antimicrobial Peptides (APD), and the Database of Research on Antimicrobial Peptides (DRAMP).

1.5.1 The Database of Antimicrobial Activity and Structure of Peptides (DBAASP).

DBAASP [22] is one of the most comprehensive resources for AMP data, containing experimentally validated information on amino acid sequences, three-dimensional structures, and biological activities. A unique feature of DBAASP is its emphasis on structural data, with experimentally determined and computationally predicted 3D conformations. This makes the database especially valuable for structure-based modeling approaches, such as molecular docking, quantitative structure–activity relationship (QSAR) models, and machine learning techniques that incorporate physicochemical features. For researchers aiming to predict AMP functionality, the detailed activity profiles in DBAASP against a

wide range of targets—including Gram-positive and Gram-negative bacteria, fungi, and cancer cells—are indispensable.

1.5.2 The Antimicrobial Peptide Database (APD).

The APD [23] is one of the earliest and most frequently used AMP databases. It was initially developed to catalog natural AMPs and their sources, including plants, insects, amphibians, mammals, and microorganisms. Over time, the database has expanded to include synthetic and modified peptides, broadening its applicability in drug design and peptide engineering. A hallmark of APD is its comprehensive classification system, where peptides are organized by source organism, structural motif, and functional category. This allows comparative studies and facilitates hypothesis generation about structure–function relationships. APD is often the first entry point for researchers new to AMP studies because of its user-friendly interface and its emphasis on biological context.

1.5.3 The Database of Research on Antimicrobial Peptides (DRAMP).

DRAMP [24] provides a large-scale and curated collection of both natural and synthetic AMPs, accompanied by detailed information about experimental conditions and references. DRAMP distinguishes itself through its dual emphasis on therapeutic application and experimental reproducibility, making it particularly valuable for translational studies. It integrates activity data with structural and clinical trial information, thus bridging the gap between basic peptide science and drug development. In addition, DRAMP’s annotation of failed or inactive peptides provides important negative data, which is especially critical for machine learning model training, where balanced datasets are required.

1.5.4 Relevance to computational prediction.

Together, DBAASP, APD, and DRAMP provide the backbone for computational peptide research. By combining sequence, activity, and structural data, these databases enable feature engineering and model training for predictive tasks. In this thesis, all three resources were utilized to compile a representative dataset of AMPs with diverse targets, serving as the foundation for training and comparing machine learning algorithms. The diversity and complementarity of these databases ensure that predictions are grounded in the widest possible experimental evidence, thus increasing model robustness and generalizability.

1.6 Designing alternative drugs

The global rise of antimicrobial resistance (AMR) poses one of the greatest threats to human health in the 21st century. Conventional antibiotics, once considered the cornerstone of infectious disease management, are increasingly losing efficacy due to the rapid evolution and dissemination of resistant pathogens [25]. This alarming trend has created an urgent need for new therapeutic strategies that not only bypass existing resistance mechanisms but also reduce the likelihood of new ones emerging. In this context, antimicrobial peptides (AMPs) have gained prominence as promising alternatives to classical antibiotics, offering unique mechanisms of action and therapeutic potential.

1.6.1 AMPs as candidates beyond conventional antibiotics.

Unlike traditional antibiotics, which often target specific bacterial processes such as cell wall synthesis, protein synthesis, or DNA replication, AMPs typically exert their activity by interacting with microbial membranes. Their amphipathic and cationic structures allow them to bind preferentially to negatively charged microbial membranes, leading to disruption, pore formation, or complete lysis of the target cell (Mahlpuu et al., 2016). Because this mechanism is relatively non-specific and physical in nature, bacteria face significant challenges in developing resistance. This feature positions AMPs as robust alternatives to small-molecule antibiotics, especially against multidrug-resistant organisms.

1.6.2 Expanding therapeutic roles.

Beyond their direct antimicrobial activity, AMPs also exhibit immunomodulatory, anti-inflammatory, and anticancer properties. They can modulate host immune responses by stimulating chemokine release, promoting wound healing, and enhancing barrier integrity. Additionally, certain AMPs display selective cytotoxicity against cancer cells, often by exploiting differences in membrane composition between malignant and healthy cells [26]. These diverse roles broaden the therapeutic relevance of AMPs, suggesting their potential utility not only as antibacterial agents but also as multi-functional drugs in oncology, virology, and immunology.

1.6.3 Design and optimization strategies.

Despite their potential, native AMPs often face limitations such as proteolytic instability, cytotoxicity, and high production costs. To address these challenges, a variety of design and optimization approaches are being pursued. Rational design strategies, such as amino acid substitution, sequence truncation, and cyclization, are used to enhance stability and specificity while reducing toxicity [9]. In parallel, computational approaches, including molecular dynamics simulations, QSAR modeling, and machine learning, have

become indispensable for designing optimized peptide sequences with improved pharmacological properties. Advances in solid-phase peptide synthesis and recombinant production technologies are also contributing to making AMPs more cost-effective and scalable.

1.6.4 AMPs in clinical development.

Several AMP-based therapeutics have progressed into clinical trials, underscoring their translational potential. Notable examples include pexiganan (a synthetic analog of magainin), omiganan (an indolicidin derivative), and LL-37 derivatives, all of which have been evaluated for applications ranging from wound healing to skin infections [6]. Although none have yet achieved widespread clinical approval, these developments highlight the progress and challenges in moving AMPs from bench to bedside. Key hurdles include optimizing therapeutic indices, ensuring stability in vivo, and overcoming delivery barriers.

1.6.5 Implications for future drug discovery.

Designing alternative drugs based on AMPs represents a paradigm shift in antimicrobial therapy. By combining natural evolutionary templates with modern computational and synthetic tools, researchers can develop peptides tailored for specific applications, including multi-class activity against bacteria, fungi, viruses, and cancer cells. In the context of this thesis, where computational models are applied to predict AMP activity, these drug design strategies underline the broader relevance of predictive modeling: by accurately identifying peptide functionality, it becomes possible to accelerate the discovery pipeline and focus resources on the most promising candidates.

1.7 Numerical application

The study of antimicrobial peptides (AMPs) has increasingly shifted from purely experimental biology toward computational and numerical approaches. This transition is largely driven by the vast and rapidly growing amount of sequence and functional data available in AMP databases, combined with the need to accelerate peptide discovery and reduce the cost of laboratory screening. Computational methods allow researchers to identify patterns, derive predictive features, and simulate peptide–target interactions before moving to experimental validation.

One of the most widely adopted computational frameworks is Quantitative Structure–Activity Relationship (QSAR) modeling, which establishes mathematical relationships between peptide descriptors and biological activity. QSAR models have been employed to evaluate how structural or physicochemical properties such as hydrophobicity, net charge, and amphipathicity correlate with antimicrobial potency. Recent advancements incorporate nonlinear methods, particularly machine learning algorithms, to im-

prove predictive accuracy and handle high-dimensional datasets where traditional regression may fail [27].

Alongside QSAR, bioinformatics pipelines play a crucial role in numerical AMP research. Sequence alignment tools, motif detection algorithms, and clustering techniques enable the classification of AMPs into functional families, which can then be linked to antimicrobial spectrum (Gram-positive, Gram-negative, fungal, viral, or anticancer). Structural modeling, ranging from homology-based 3D prediction to molecular dynamics simulations, provides additional layers of insight into mechanism of action, such as membrane disruption or intracellular targeting.

In recent years, machine learning and deep learning approaches have revolutionized numerical applications in AMP research. Techniques such as support vector machines (SVMs), random forests, and convolutional neural networks (CNNs) have been applied to AMP classification, prediction of target specificity, and activity spectrum differentiation. For example, CNNs can process raw sequence data directly, learning complex representations that surpass hand-engineered descriptors, whereas ensemble methods like random forests integrate multiple features for robust prediction. These numerical approaches make it feasible to predict AMP functions across multiple classes (e.g., Gram-positive vs. Gram-negative bacteria, fungi, viruses, and cancer cells), a task that would be prohibitively expensive to address exclusively through wet-lab assays.

Furthermore, the integration of numerical simulations with experimental feedback is becoming a defining feature of modern AMP discovery. Iterative cycles of computational prediction, peptide synthesis, and biological testing allow refinement of models and improvement in predictive performance. This synergy accelerates the development of AMPs as viable therapeutic alternatives, particularly in the context of rising antibiotic resistance. By enabling large-scale virtual screening, numerical applications significantly reduce the experimental burden and provide a foundation for rational AMP design.

1.8 Feature engineering

Feature engineering is a central step in computational biology and particularly critical for predicting the functional properties of antimicrobial peptides (AMPs). The raw peptide sequence provides a starting point, but in order to build predictive machine learning (ML) models, it must be transformed into quantitative descriptors that capture relevant biological and chemical information. In this thesis, three main categories of features were utilized: amino acid composition, physicochemical properties, and evolutionary/biological context.

1.8.1 Amino acid composition.

One of the most straightforward yet informative ways to represent peptides is by quantifying the relative abundance of each amino acid. This approach captures global sequence-level characteristics without requiring sequence alignment or motif detection. Percentages of amino acid content are especially useful for identifying trends such as enrichment of positively charged residues (lysine, arginine, histidine) that facilitate electrostatic interactions with negatively charged bacterial membranes, or prevalence of hydrophobic residues (leucine, isoleucine, phenylalanine) that enable membrane insertion and disruption [28]. While amino acid composition cannot fully describe sequence order, it provides a baseline representation that has been shown to perform competitively in machine learning tasks for AMP prediction.

1.8.2 Physicochemical properties.

Beyond amino acid counts, peptides can be described through aggregate physicochemical features that directly relate to their antimicrobial function. Properties such as net charge, hydrophobicity, molecular weight, isoelectric point, Boman index (protein-binding potential), and aliphatic index provide numerical insights into how peptides interact with microbial membranes or intracellular targets [29]. For example, cationic charge is strongly correlated with bacterial membrane disruption, while hydrophobicity governs the ability of peptides to penetrate lipid bilayers. Combining multiple physicochemical descriptors allows models to capture complex relationships between peptide structure and antimicrobial activity. Moreover, these features are scale-invariant and can be easily standardized, making them well-suited for ML workflows.

1.8.3 Evolutionary and biological context features.

A third dimension of feature engineering involves higher-level biological descriptors derived from the evolutionary origin of peptides. In this work, metadata such as whether the peptide originates from cellular organisms (bacteria, plants, animals) or acellular sources (viruses), and whether the source species possesses tissue specialization, were incorporated. These features provide evolutionary signals that complement sequence-derived properties. For example, peptides from multicellular organisms often serve regulatory or immune defense functions, whereas microbial peptides may have evolved primarily as competitive weapons. Incorporating such contextual features has been shown to improve prediction accuracy, as they provide functional priors not directly inferable from sequence alone [30][31].

1.8.4 Integrative representation.

The combination of amino acid composition, physicochemical descriptors, and evolutionary metadata allows for a richer and more holistic representation of peptides. While individual feature categories capture important biological aspects, their integration enables models to distinguish between subtle functional classes—for example, peptides active against Gram-positive versus Gram-negative bacteria, or those targeting fungi, viruses, or cancer cells. Feature engineering thus forms the foundation of predictive modeling, determining the quality of the input space upon which machine learning methods such as Naïve Bayes, Random Forests, Support Vector Machines, and Convolutional Neural Networks can operate.

In summary, feature engineering transforms raw biological sequences into structured numerical data suitable for computational learning. The design choices in feature construction not only influence model performance but also reflect underlying biological hypotheses about what makes antimicrobial peptides effective. By combining compositional, physicochemical, and evolutionary descriptors, this thesis seeks to maximize predictive accuracy while maintaining interpretability of the models.

1.9 Feature selection

While feature engineering determines how biological sequences are transformed into numerical descriptors, feature selection addresses the challenge of reducing redundancy and retaining only the most informative variables. This step is critical because the initial feature space, particularly when amino acid composition, physicochemical properties, and evolutionary descriptors are combined, can be very high-dimensional. High-dimensional feature spaces increase computational costs, exacerbate the risk of overfitting, and often contain correlated or irrelevant features that dilute predictive performance [32]. By systematically identifying the most discriminative features, feature selection improves both the efficiency and accuracy of machine learning models for antimicrobial peptide (AMP) prediction.

1.9.1 The need for feature selection in AMP prediction.

In peptide bioinformatics, descriptors such as amino acid frequencies, dipeptide compositions, hydrophobicity scales, charge indices, and evolutionary information can result in hundreds or even thousands of potential features. Many of these are not equally relevant for distinguishing AMPs with specific targets (e.g., Gram-positive vs. Gram-negative bacteria, fungi, viruses, or cancer cells). Moreover, correlated features—such as hydrophobicity and aliphatic index—may carry overlapping information, which can bias the learning

algorithm. Feature selection reduces this redundancy, allowing models to focus on features that best capture biological variability [33].

1.9.2 Filter-based methods.

Filter methods rank features according to statistical criteria that measure their association with the target variable. Examples include mutual information, chi-squared tests, Pearson correlation, and analysis of variance (ANOVA). These methods are computationally efficient and independent of the chosen classifier. For AMP prediction, filter methods have been used to identify key amino acid preferences (e.g., enrichment of arginine and lysine in cationic AMPs) or structural properties such as net charge and amphipathicity [30]. Although filter methods may ignore interactions between features, they provide a useful first step in dimensionality reduction.

1.9.3 Wrapper-based methods.

Wrapper approaches evaluate subsets of features by training and testing the predictive model on different feature combinations. Techniques such as recursive feature elimination (RFE), forward selection, and backward elimination fall into this category. In peptide classification tasks, wrapper methods have proven effective in identifying compact subsets of descriptors that maximize classification accuracy (Porto et al., 2018). However, these methods are computationally more expensive, especially when applied to large datasets with many features.

1.9.4 Embedded methods.

Embedded feature selection integrates the selection process within model training. Regularization-based approaches, such as Lasso (L1 penalty) and Elastic Net, automatically shrink less informative coefficients toward zero, thereby performing implicit feature selection. Tree-based methods like Random Forests and Gradient Boosting naturally rank feature importance by measuring information gain or Gini index reduction. In the context of AMPs, embedded methods are particularly useful since they can highlight the physicochemical features most responsible for antimicrobial activity while simultaneously training the model [34].

1.9.5 Dimensionality reduction techniques.

Another class of methods involves transforming the feature space into a lower-dimensional representation rather than directly selecting features. Principal Component Analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE), and autoencoders are frequently employed for this purpose. PCA, for instance, can condense correlated descriptors into

principal components while retaining the majority of variance in the data [35]. While such transformations may reduce interpretability, they are effective in simplifying complex peptide datasets.

1.9.6 Balancing interpretability and predictive power.

A key consideration in AMP research is the trade-off between model interpretability and predictive performance. While dimensionality reduction methods may enhance accuracy, they obscure direct biological meaning. On the other hand, feature selection methods that retain specific amino acid or physicochemical descriptors preserve interpretability, which is crucial for generating biological hypotheses and guiding peptide design. This thesis employs a combination of filter-based analysis to reduce redundancy and embedded methods to identify biologically meaningful features, ensuring that the models remain both efficient and explainable.

In summary feature selection is an indispensable component of AMP prediction pipelines, complementing feature engineering by refining the input space into a more compact and informative set. By removing redundancy, highlighting discriminative variables, and improving generalization, feature selection directly influences the performance of downstream models such as Naïve Bayes, Random Forests, Support Vector Machines, and Convolutional Neural Networks. In this work, it is treated not merely as a preprocessing step but as a means of integrating biological insight into computational modeling.

1.10 Machine Learning Methods

Machine learning (ML) methods have become central to the task of predicting antimicrobial peptide (AMP) activity and specificity. The intrinsic diversity of peptide sequences, combined with their structural and functional complexity, makes this a challenging classification problem. Traditional biochemical heuristics alone are insufficient to capture the nonlinear dependencies between sequence composition, physicochemical properties, and biological activity. ML algorithms provide the necessary computational framework to learn such relationships from large datasets and generalize predictions to novel peptides.

In this thesis, five machine learning methods are applied to predict AMP functionality across five target classes—Gram-positive bacteria, Gram-negative bacteria, fungi, viruses, and cancer cells. The models investigated include Naïve Bayes (NB), Random Forests (RF), Multi-layer Perceptron (MLP), Support Vector Machines (SVM), and Convolutional Neural Networks (CNN). Each of these algorithms brings unique strengths in handling high-dimensional biological data and has been employed in prior studies of peptide classification.

1.10.1 Naïve Bayes Classifier

Naïve Bayes is a probabilistic classifier grounded in Bayes' theorem, assuming conditional independence between features given the class. Despite the simplicity of this assumption, NB has demonstrated effectiveness in text classification, genomics, and peptide activity prediction [36]. In the AMP context, NB can model the relationship between amino acid composition, physicochemical descriptors, and antimicrobial activity by estimating class-conditional probability distributions. Its advantages include computational efficiency, robustness to small datasets, and interpretability. However, NB may underperform when features are highly correlated, a common situation in peptide descriptors such as hydrophobicity and charge.

1.10.2 Random Forests

Random Forests, an ensemble learning method based on decision trees, are widely used in bioinformatics because of their ability to handle nonlinear relationships, mixed data types, and noisy features (Breiman, 2001). In peptide prediction tasks, RFs construct multiple decision trees using bootstrap samples and random feature subsets, and aggregate their outputs through majority voting. This reduces overfitting compared to single decision trees and provides an inherent measure of feature importance. RFs have been shown to capture subtle sequence-activity patterns in AMPs, highlighting discriminative physicochemical features such as charge density, hydrophobic moment, and residue distribution [37]. Their interpretability and robustness make them a strong baseline for peptide classification.

1.10.3 Multi-layer Perceptron (MLP).

The Multi-layer Perceptron is a feed-forward neural network capable of learning nonlinear mappings between input features and outputs. By applying nonlinear activation functions across hidden layers, MLPs can model complex dependencies between amino acid composition, evolutionary descriptors, and AMP activity [38]. In this work, MLPs are trained on engineered features including amino acid frequencies, physicochemical indices, and biological source annotations. While MLPs require careful tuning of hyperparameters (e.g., number of layers, hidden units, and learning rates), they provide flexibility in approximating arbitrary decision boundaries. MLPs also benefit from regularization strategies such as dropout, which help mitigate overfitting in high-dimensional feature spaces.

1.10.4 Support Vector Machines (SVM).

Support Vector Machines are powerful classifiers that find the optimal hyperplane separating classes in a transformed feature space. By employing kernel functions such as radial basis function (RBF) or polynomial kernels, SVMs can project peptide descriptors into higher dimensions, allowing for the separation of classes that are not linearly separable in the original space [39]. SVMs have consistently achieved state-of-the-art performance in peptide classification tasks, especially when sample sizes are limited [30]. Their strength lies in maximizing the margin between classes, which enhances generalization ability. However, SVMs can be computationally intensive with large datasets and may require careful kernel selection and parameter tuning.

1.10.5 Convolutional Neural Networks (CNN).

Convolutional Neural Networks have revolutionized sequence analysis by learning local motifs and higher-order patterns directly from raw sequence representations. Unlike traditional models that depend heavily on feature engineering, CNNs can automatically detect relevant motifs such as amphipathic patterns or charge clusters within peptide sequences [40]. In AMP prediction, CNNs treat amino acid sequences as ordered vectors or one-hot encoded matrices, applying convolutional filters to extract features indicative of antimicrobial activity. This makes CNNs particularly suitable for capturing local sequence dependencies and higher-level abstractions. Despite requiring larger datasets and higher computational resources, CNNs represent a cutting-edge approach that aligns with modern deep learning practices in bioinformatics.

1.10.6 Model Selection in AMP Prediction

The diversity of algorithms applied here reflects a balance between interpretability and predictive power. Simpler methods such as NB and RF provide transparent models with interpretable feature contributions, making them suitable for hypothesis generation and biological validation. More complex models like MLPs, SVMs, and CNNs capture non-linear relationships and sequence-level patterns at the cost of increased computational complexity and reduced interpretability. The use of cross-validation ensures robust evaluation of generalization across target classes, and the comparative analysis highlights how different ML paradigms handle the unique challenges of AMP prediction.

The application of diverse machine learning methods provides complementary perspectives on AMP classification. NB offers simplicity and interpretability, RF balances predictive performance with feature importance insights, MLP and SVM model non-linear dependencies in engineered features, and CNN leverages raw sequences for automatic motif extraction. Together, these models provide a comprehensive framework for under-

standing the computational prediction of AMP targets. In this thesis, their performance is systematically compared across multiple datasets and feature sets, contributing to the growing body of research on data-driven peptide therapeutics.

1.11 Model Evaluation

Model evaluation is a crucial stage in any machine learning pipeline, as it determines the reliability, robustness, and generalizability of predictive models. In the context of antimicrobial peptide (AMP) prediction, evaluation is particularly important because of the biological and clinical implications of classification errors. Misclassifying peptides as active against a pathogen when they are not, or vice versa, may misguide downstream experimental validation and drug development efforts. Therefore, rigorous evaluation strategies are necessary to ensure that the models not only perform well on the training data but also generalize effectively to unseen peptides with diverse physicochemical and evolutionary properties (Witten et al., 2017).

1.11.1 Cross-Validation Approaches

Cross-validation remains the gold standard for model assessment in bioinformatics. It allows for efficient use of limited data and provides an estimate of predictive performance under realistic conditions. In this thesis, k-fold cross-validation was employed, in which the dataset is divided into k subsets of approximately equal size. Each model is trained on k-1 folds and tested on the remaining fold, cycling through all partitions to obtain an averaged performance score [41]. This reduces bias that may arise from a single train-test split and helps mitigate overfitting, particularly important for peptide datasets that often have imbalanced distributions across functional classes.

Stratified cross-validation, which preserves class proportions in each fold, is especially useful when predicting multiple targets such as Gram-positive bacteria, Gram-negative bacteria, fungi, viruses, and cancer cells. Without stratification, underrepresented classes (e.g., antiviral AMPs) may be omitted in some folds, skewing evaluation results [42].

1.11.2 Evaluation Metrics

Different metrics provide complementary insights into model performance:

- Accuracy measures the proportion of correctly classified peptides across all classes. While intuitive, accuracy may be misleading in imbalanced datasets where one class dominates.
- Precision, Recall, and F1-score provide a more nuanced view. Precision indicates how many peptides predicted to target a certain organism actually do so, whereas

recall measures how many true peptides were correctly identified. The F1-score harmonizes both, offering a balanced assessment [43].

- Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC) evaluate the trade-off between sensitivity and specificity across thresholds. ROC-AUC is widely adopted in peptide prediction to quantify discriminative ability independent of classification threshold [44].
- Precision-Recall (PR) Curves are particularly useful for imbalanced multi-class problems, as they highlight performance on minority classes where false negatives may be more detrimental than false positives.

For multi-class classification problems such as those in this thesis, macro- and micro-averaged F1-scores are employed. Macro-averaging treats all classes equally, while micro-averaging weights them by frequency, ensuring that both minority and majority classes are adequately represented in evaluation.

1.11.3 Overfitting and Generalization

A critical challenge in machine learning applied to biological data is overfitting. Complex models, particularly deep learning architectures such as Convolutional Neural Networks (CNNs), can achieve high training accuracy but fail to generalize. To address this, cross-validation is combined with regularization techniques, dropout layers, and early stopping criteria [45]. Model evaluation therefore extends beyond metrics, encompassing careful monitoring of learning dynamics and validation error to ensure robust performance.

1.11.4 Comparison Across Methods

Because multiple models are considered in this work (Naïve Bayes, Random Forest, Support Vector Machine, Multi-Layer Perceptron, and CNN), evaluation is also comparative. Each algorithm has unique strengths: Naïve Bayes provides transparency but may underperform on correlated features, Random Forests yield robust feature importance insights, SVMs excel in handling high-dimensional descriptors, while neural networks can capture non-linear relationships but require larger datasets. Through standardized evaluation metrics and cross-validation, the relative advantages and limitations of each method become clearer.

1.11.5 Relevance to Biological Interpretation

Finally, evaluation metrics are not merely numerical indicators, but have biological significance. The high recall in the identification of anticancer peptides, for instance, suggests greater utility in therapeutic development where missing candidates would be

costly. Conversely, high precision in the prediction of antibacterial AMPs ensures fewer false positives, saving resources in experimental verification. Thus, the choice of evaluation strategy directly influences how computational predictions can be trusted in guiding peptide discovery pipelines.

1.11.6 Concluding Remarks

Model evaluation represents the bridge between computational prediction and biological application. By employing cross-validation, diverse metrics, and strategies to avoid overfitting, one ensures that models are both statistically sound and biologically meaningful. In this paper, the evaluation framework is designed to balance predictive accuracy with interpretability and generalizability, enabling fair comparisons between algorithms and establishing a reliable foundation for functional AMP prediction.

1.12 Multi-class prediction

Machine learning problems in computational biology often extend beyond binary classification, where the task is to distinguish between two categories (e.g., antimicrobial vs. non-antimicrobial peptides). In antimicrobial peptide (AMP) research, a more complex challenge arises: peptides can exhibit diverse and overlapping functional activities against multiple biological targets. For instance, a single peptide may be active against Gram-positive bacteria, Gram-negative bacteria, fungi, viruses, or even cancer cells. This inherently multiclass prediction problem requires models capable of handling multiple outcomes simultaneously while preserving biological interpretability [46].

In binary AMP prediction, the focus is often on distinguishing AMPs from non-AMPs. While useful, this formulation oversimplifies the biological reality. Many AMPs are multifunctional, displaying activities against different classes of microbial or cell types. Predicting these activities requires extending the classification framework to account for multiple classes. There are two strategies:

- Single-label multiclass classification; each peptide is assigned to exactly one functional category. This is appropriate when peptides are known to exhibit a dominant activity, such as targeting Gram-positive bacteria only.
- Multilabel classification; peptides may simultaneously belong to more than one class (e.g. antibacterial and anticancer). Although more realistic, this approach is computationally more complex, requiring models to handle overlapping class memberships [47].

In this thesis, the focus is on single-label multiclass classification, where each peptide is predicted to belong to one of five functional categories: Gram-positive, Gram-negative,

fungi, viruses, or cancer. Each category represents harmful activity towards itself, type of microbe they represent.

1.12.1 Challenges in Multi-class Prediction

Several methodological and biological challenges arise.

Class inequality; AMP datasets are often skewed, with antibacterial peptides (Gram-positive and Gram-negative) being far more abundant than antiviral or anticancer peptides. This imbalance can bias models toward the majority classes, reducing predictive accuracy for minority classes [48].

Feature overlap; physicochemical descriptors (e.g., charge, hydrophobicity) may not be uniquely discriminative across classes. For example, cationicity contributes to both antibacterial and anticancer activity, complicating model decision boundaries.

Error propagation; misclassifying peptides in multiclass tasks introduces more severe errors compared to binary classification, as the model must choose between several possible incorrect categories.

1.12.2 Machine Learning Strategies

Classical machine learning models handle multiclass problems using different approaches:

One-vs-Rest – the model learns a binary classifier for each class against all others, with the highest-confidence prediction assigned as output. Support Vector Machines (SVM) and Naïve Bayes classifiers commonly use this strategy [49].

One-vs-One (OvO) – the model trains classifiers for each pair of classes, and predictions are made by majority voting. This is computationally more expensive but often improves accuracy.

Native multi-class models – decision tree ensembles (Random Forests) and neural networks (MLPs, CNNs) naturally extend to multi-class outputs without explicit decomposition.

Deep learning methods such as Convolutional Neural Networks have shown particular promise for multi-class AMP prediction, as they can capture non-linear relationships and learn hierarchical representations of sequence features [38].

1.12.3 Evaluation of Multi-class Models

Evaluation in multi-class prediction requires metrics that account for multiple categories simultaneously:

Macro-averaged Precision, Recall, and F1-score treat all classes equally, regardless of their frequency. This highlights performance on minority classes (e.g., antiviral peptides).

Micro-averaged metrics weight results by class size, providing a measure that reflects overall dataset balance.

Confusion matrices are especially useful, as they display misclassification patterns and help identify where models systematically confuse classes (e.g., Gram-positive vs. Gram-negative).

Multi-class ROC curves and AUC can also be extended, although interpretation becomes more complex.

1.12.4 Biological Significance

Multi-class prediction is not only a computational challenge but also a biological necessity. Many AMPs are pleiotropic, meaning they perform multiple roles depending on context. Even in single-label formulations, predicting functional categories offers valuable insights into the mechanisms of peptide activity. Correctly identifying whether a peptide targets bacterial pathogens versus cancer cells can guide drug discovery pipelines, resource allocation, and therapeutic prioritization [50].

The transition from binary to multi-class prediction reflects a growing emphasis on biological realism in AMP computational research. By addressing challenges such as class imbalance, overlapping features, and evaluation complexity, machine learning methods can provide more accurate and meaningful predictions. In this thesis, the implementation of multi-class prediction allows for a nuanced comparison of classical and deep learning models, highlighting both methodological strengths and biological implications in peptide functional classification.

Chapter 2

Methods

2.1

Chapter 3

[Chapter title]

References to book [51], [**bib:1**] scientific papers in journals [52], conference papers [53], and web pages [54].

Equations should be numbered:

$$y = \frac{\partial x}{\partial t} \tag{3.1}$$

Chapter 4

[Chapter title]

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4.1 [Section title]

4.2 [Subsection title]

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Each table in the document should be referred to at least once (Tab. 4.1).

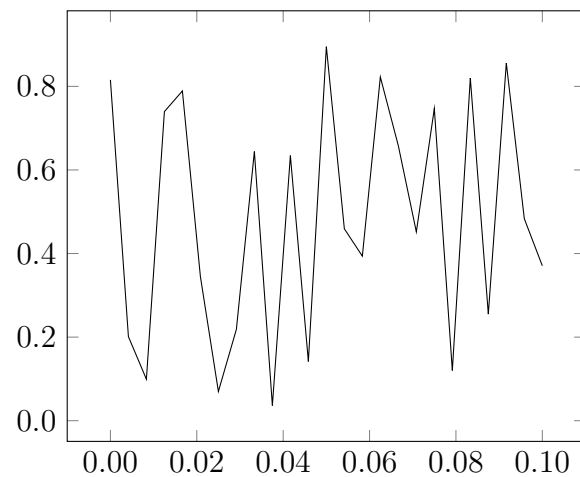


Figure 4.1: Figure caption.

Table 4.1: A caption of a table is ABOVE it.

ζ	method						
	alg. 1	alg. 2	alg. 3			alg. 4, $\gamma = 2$	
			$\alpha = 1.5$	$\alpha = 2$	$\alpha = 3$	$\beta = 0.1$	$\beta = -0.1$
0	8.3250	1.45305	7.5791	14.8517	20.0028	1.16396	1.1365
5	0.6111	2.27126	6.9952	13.8560	18.6064	1.18659	1.1630
10	11.6126	2.69218	6.2520	12.5202	16.8278	1.23180	1.2045
15	0.5665	2.95046	5.7753	11.4588	15.4837	1.25131	1.2614
20	15.8728	3.07225	5.3071	10.3935	13.8738	1.25307	1.2217
25	0.9791	3.19034	5.4575	9.9533	13.0721	1.27104	1.2640
30	2.0228	3.27474	5.7461	9.7164	12.2637	1.33404	1.3209
35	13.4210	3.36086	6.6735	10.0442	12.0270	1.35385	1.3059
40	13.2226	3.36420	7.7248	10.4495	12.0379	1.34919	1.2768
45	12.8445	3.47436	8.5539	10.8552	12.2773	1.42303	1.4362
50	12.9245	3.58228	9.2702	11.2183	12.3990	1.40922	1.3724

Chapter 5

Experiments

Chapter 6

Summary

- synthetic description of performed work
- conclusions
- future development, potential future research
- Has the objective been reached?

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Appendices

Technical documentation

List of abbreviations and symbols

AMP antimicrobial peptides

ML machine learning

DNA deoxyribonucleic acid

MVC model–view–controller

N cardinality of data set

μ membership function of a fuzzy set

$\mathbb{E}$ set of edges of a graph

$\mathcal{L}$ Laplace transformation

List of additional files in electronic submission (if applicable)

Additional files uploaded to the system include:

- source code of the application,
- test data,
- a video file showing how software or hardware developed for thesis is used,
- etc.

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